

ROBOTICS AND AI

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Class X

Assignment 5–7: Python Programming

Due Date: 22.09.26

35 marks

INSTRUCTIONS

- Don't copy from AI apps
 - Write and test each program before submission; attach a screenshot of the output where applicable
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Assignment 5

Python: Lists

Question A1: Write a Python program to input a list of numbers and display: (4 marks)

- The largest number
- The smallest number
- The sum of all numbers
- The average of the numbers

Question A2: Write a Python program to create a list of numbers and display a new list containing only the even numbers from the original list. (3 marks)

Question A3: Write a Python program to create a list of numbers and remove all duplicate elements from it, displaying the resulting list. (3 marks)

Question A4: Which of the following methods is used to add an element at the end of a list? (1 mark)

Note: Choose the right option.

- | | |
|--------------|--------------|
| (a) insert() | (b) add() |
| (c) append() | (d) extend() |

Question A5: What is the key difference between a list and a tuple in Python? Give one example each. (2 marks)

Assignment 6

Python: Tuples

Question B1: Write a Python program to create a tuple of numbers and: (5 marks)

- Count the number of even elements

- Find the maximum and minimum elements
- Display the tuple in reverse order

Question B2: Which of the following statements about tuples is TRUE? (1 mark)

Note: Choose the right option.

- | | |
|---|--|
| (a) Tuples are mutable | (b) Tuples are immutable |
| (c) Tuples can be modified after creation | (d) Tuples cannot store duplicate values |

Question B3: Write a Python program to convert a given list into a tuple, and then convert that tuple back into a list. (2 marks)

Question B4: What are the advantages of tuple over list? (2 marks)

Assignment 7

Python: Strings

Question C1: Write a Python program to input a string and display: (4 marks)

- Number of vowels
- Number of consonants
- Number of digits
- Number of spaces

Question C2: Write a Python program to input a sentence and create a list containing all the words having more than 4 characters. Display the list and its length. (3 marks)

Question C3: Which of the following best describes strings in Python? (1 mark)

Note: Choose the right option.

- | | |
|------------------------------------|--------------------------------------|
| (a) Mutable sequence of characters | (b) Immutable sequence of characters |
| (c) Ordered collection of numbers | (d) None of these |

Question C4: Write a Python program to reverse a given string without using slicing or any built-in reverse function. (2 marks)

Question C5: Write a Python program to input a string and a substring, and count the number of times the substring occurs in the string. (2 marks)